An N-gram model figures out how likely a word is to come after a sentence.

N decides how long the sentence / context / prefix is.

bad Things can happen

[0] [1] [2] [3]

a 3-gram model would calculate:

$$P(\text{happen} \mid \text{Things can}) = P(w_3 \mid w_1^2) = P(w_3 \mid w_{3-(3-1)}^{3-1})$$

$$= P(w_n \mid w_{n-(N-1)}^{n-1})$$

an N-gram model works with (N-1)-gram context.

$$P(w_n \mid w_{n-(N-1)}^{n-1}) = \frac{C(w_{n-(N-1)}^{n})}{C(w_{n-(N-1)}^{n-1})}$$

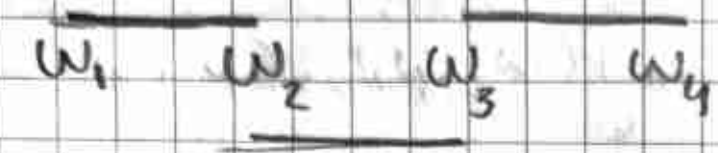
(count)

N-GRAMS



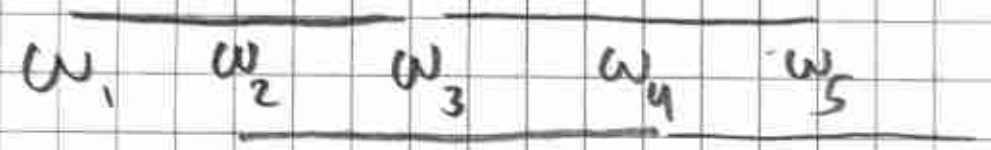
UNI-GRAM

2



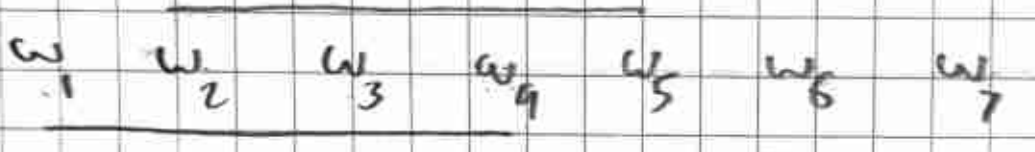
BI-GRAM

3



TRI-GRAM

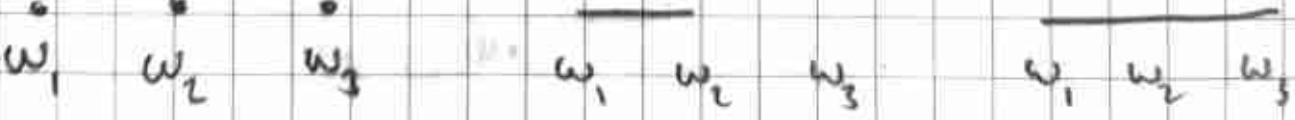
4



I am happy because I am learning

A sentence is a bunch of word sequences stitched together. They each carry meaning that's less than the entire sentence, but more than each individual word.

(individual words) (bi-grams) (entire sentence)



Give the least context. Give more context. Give the entire context.

P(The teacher drinks tea) = P(the) P(teacher | the)
P(drinks | the teacher) P(tea | the teacher drinks)

$P(w_1\ w_2\ w_3) = P(w_1)\ P(w_2|w_1)\ P(w_3|w_1\ w_2)$

$\simeq P(the)\ P(teacher|the)\ P(drinks|teacher)\ P(tea|drinks)$

• The teacher drinks tea ⇒ This requires looking for 4-grams.

Can be approximated by ...

• The teacher drinks tea ⇒ This only requires looking for bi-grams.

Since we're way more likely to find bi-grams than n-grams, we get meaningful probabilities by using the approximation that looks for bi-grams only.

(!) won't we get a better approximation if we worked with tri-grams instead?

P(The teacher drinks tea) ≈

<s> <s> The teacher drinks tea.

This is a better estimate, but you now need to find tri-grams.

Work with more n-grams

↳ get a better approximation.

overdo it

↳ risk not finding the "longer" n-gram in your corpus.

... I like how this approximation works.

$$P(w_n \mid w_1^{n-1}) = \frac{P(w_1^{n-1} w_n)}{P(w_1^{n-1})} = \frac{C(w_1^n)}{C(w_1^{n-1})}$$

$C(w_1^n) \Rightarrow$ n-gram

$C(w_1^{n-1}) \Rightarrow$ (n-1)-gram

(Prefix + end) (Prefix)

an n-gram model, looks for n and n-1 grams.

it's job is to find $P(\text{end} \mid \text{Prefix}) = P(w_n \mid w_{1+n-N}^{n-1})$

$P(w_6 \mid w_4^5)$

1 2 3 4 5 6 $n=6$, $N=3$ $n-(N-1)$